2330 Dashboard — System Architecture & How It Works

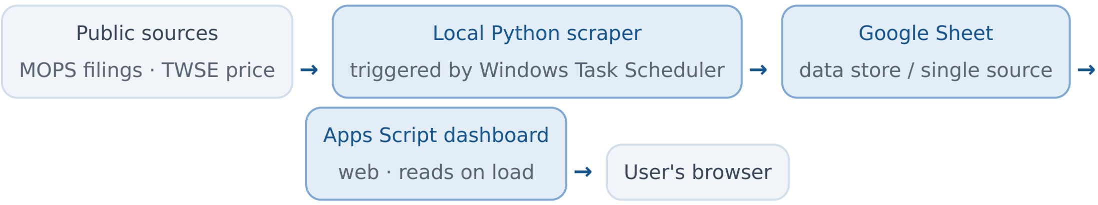
A live monitoring dashboard built entirely on public data · zero cost · Google Apps Script + local Python

Sources: MOPS (Market Observation Post System) + TWSE public APIs · Independent, unofficial visualization

1 · Overview

Turns TSMC (2330) **public data** into a web dashboard: recent material-information filings, share price and a 12-month average, plus an AI chat you can ask about the filings in plain language. Public data only — no confidential data, no self-hosted server.

2 · Architecture & data flow



- **Filings:** a program pulls from MOPS (per-company, monthly `ajax_t05st01` — the authoritative feed), parses to a table (date / subject / clause / full description), and writes to a Google Sheet.
- **Price:** fetched live from TWSE public APIs (STOCK_DAY daily close / FMSRFK monthly average) on each page load.
- **AI Q&A:** the dashboard connects Google Gemini to answer natural-language questions about the filings (see §4).

3 · Update mechanism (three roles + read model)

- **Windows Task Scheduler (local — the trigger):** runs the scraper every weekday at 08:40. **Why local:** MOPS is only reachable from a local network (blocked from cloud), so the scraper and its schedule live on the local machine.
- **Local Python (the fetcher):** scrapes new MOPS filings → appends them to the Sheet → stamps a "verified-through" date (advances even on days with no new filing).
- **Apps Script dashboard (the presenter):** runs once per page open / refresh, reading the Sheet and TWSE **at that moment**. No background push, no auto-refresh — reload to update.
- **Price frequency:** the source is daily trading data, so it changes roughly once per trading day (after settlement), not tick-by-tick intraday. "Live" means the fetch happens on load.

4 · AI Q&A (Google Gemini)

- Apps Script calls the Gemini API **server-side**; the API key and a passcode are kept in **Script Properties** (not hard-coded). A passcode gate protects the chat.
- **Grounding:** the most recent ~40 filings (including the full description) from the Sheet — so "what did this filing say" is answered from the actual scraped MOPS data.

> **Limitation:** clause explanations (e.g. M11 = Article 4, item 11 of the TWSE material-information procedures) come from Gemini's general knowledge plus a prompt hint — **not an authoritative regulation text**. Treat as AI-assisted; defer to the official rules.

5 · Environment & dependencies

- **Local Windows** must be on and online for the schedule to run (ingestion depends on the local machine).
- **Google account + OAuth token** (to write the Sheet); the **Apps Script Web App** serves a public `/exec` URL (executes as the owner).
- **Gemini free-tier key** (Flash model, ~1,500 requests/day).
- **Data sensitivity:** all public (MOPS/TWSE), no internal data; the passcode blocks random users.

Sources: MOPS + TWSE public APIs. An independent visualization built on public data — not an official document and not investment advice.